CampusMart

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***Abstract*— This paper presents the development of a campus-based student marketplace platform designed to support secure and efficient buying, selling, and exchange of products within a college community. The system addresses common issues associated with informal student trading, such as limited product visibility, trust concerns, and communication difficulties. Developed using the Flask framework with SQLAlchemy ORM and SQLite database integration, the platform provides features including user authentication, product listing management, keyword-based search, category filtering, wishlist functionality, order tracking, and direct buyer–seller messaging. To improve product discovery, the system incorporates location-based search using distance calculation techniques and personalized recommendations through collaborative filtering. The application is deployed on a cloud-hosted environment using Gunicorn to ensure reliable accessibility and performance. The proposed platform promotes affordable student commerce, encourages product reuse, and strengthens connectivity within the campus community.**

Keywords*— *Campus Marketplace, Student Marketplace System, Flask Framework, Collaborative Filtering, Location-Based Search.

I. INTRODUCTION

Students often require access to essential goods such as textbooks, electronics, and furniture at affordable prices. However, existing online marketplaces are not tailored to campus-specific needs and often lack trust and user verification, leading to inefficiencies in student transactions. A campus-based marketplace system provides a secure and localized platform restricted to verified users within an institution. This approach enhances trust, improves transaction reliability, and promotes the reuse of goods within the student community.

This paper presents the design and implementation of a scalable backend system for a campus-based student marketplace. The system focuses on efficient data management, secure authentication, and seamless interaction through RESTful APIs, and is deployed on cloud infrastructure to ensure scalability and reliability.

1.1 Need for Campus Marketplace System

Students often depend on informal messaging groups and social platforms to buy or sell products within the campus community. However, these methods create several challenges, including difficulty in finding affordable second-hand products, lack of trust and transparency, and the absence of an organized product discovery mechanism. Communication between buyers and sellers also becomes inefficient, leading to delays and confusion during transactions. Moreover, existing online marketplace platforms are generally not campus-focused, making them less suitable for meeting the specific needs of student communities.

1.2 Role of the Proposed Platform

The proposed platform provides a centralized marketplace designed specifically for students, allowing them to buy, sell, and exchange products within the campus community. It supports secure product listing and management to ensure organized and reliable transactions. The system enables direct communication between buyers and sellers, improving interaction and reducing communication barriers. Additionally, it incorporates a recommendation system for personalized product discovery and uses location-based search to help users find nearby products more efficiently. Overall, the platform simplifies campus-level commerce by creating a secure, accessible, and student-focused trading environment.

1.3 Benefits of the Proposed System

For Students

Students can gain affordable access to a variety of products available within the campus community. Organized listings and search features support faster product discovery, helping users locate required items more efficiently. Communication between buyers and sellers becomes easier and more convenient, improving the overall transaction process. This approach also helps reduce unnecessary expenses by enabling students to purchase reasonably priced and reusable products from fellow campus members.

For Campus Community

The platform encourages product reuse by allowing students to exchange and resell items within the campus community instead of discarding them. This practice promotes sustainable consumption by reducing waste and extending the usability of products. In addition, the marketplace strengthens student interaction by creating opportunities for communication and collaboration through campus-based buying and selling activities.

Platform Benefits

The system is built with a secure architecture that supports safe user access and reliable data handling. Its scalable design allows the platform to accommodate increasing users and product listings without affecting performance. Personalized recommendations improve product discovery by suggesting relevant items based on user activity and preferences. In addition, efficient product management features help organize listings, update product information, and maintain a smooth marketplace experience.

II. LITERATURE REVIEW

Several studies have explored online marketplace systems that support peer-to-peer buying and selling through digital platforms. Traditional e-commerce applications provide broad accessibility but often lack localized interaction, trusted user verification, and community-focused product discovery, which are important in campus environments. As a result, students frequently rely on informal communication channels such as messaging groups and social media platforms for exchanging products, leading to issues related to trust, product visibility, and inefficient communication.

Recent advancements in web technologies and intelligent recommendation systems have improved the efficiency of digital marketplace platforms. Technologies such as Flask, SQLAlchemy ORM, and cloud deployment services are widely used to build secure and scalable applications. In addition, collaborative filtering techniques and location-based search mechanisms help enhance personalized product discovery and nearby product accessibility. The proposed Campus Mart system integrates these features into a centralized platform designed specifically for student communities.

2.1 Traditional Online Marketplace Systems

- Digital marketplace platforms such as OLX, Facebook Marketplace, and eBay support large-scale peer-to-peer buying and selling.
- These platforms provide features including product listings, category-based browsing, and buyer–seller communication.
- However, they are not specifically designed for student communities or campus-based trading.
- Common limitations include:
  - Lack of verified campus users
  - Large numbers of irrelevant product listings
  - Trust and safety concerns

2.2 Recommendation Systems in E-Commerce

- Modern e-commerce platforms use recommendation systems to improve user experience and engagement.
- Collaborative filtering is one of the most commonly used recommendation techniques.
- Recommendation systems analyze:
  - User behavior
  - Wishlist activity
  - Purchase history
  - Product views and interactions
- Cosine similarity methods help identify users with similar interests.
- Personalized recommendations reduce search effort and improve relevant product discovery.

2.3 Location-Based Product Discovery

- Location-aware systems are increasingly used in modern marketplace applications.
- GPS-based search helps users identify products available in nearby locations.
- Distance calculation techniques such as the Haversine formula are used to measure proximity between buyers and sellers.
- Location-based product discovery provides:
  - Faster transactions
  - Reduced travel effort
  - Easier product pickup
- Improved trust within localized communities
- These features are especially useful in campus-based marketplace environments.

III. SYSTEM ARCHITECTURE

The architecture of the proposed Campus Mart system is designed to provide a secure, scalable, and efficient platform for campus-based product trading. The system follows a client–server architecture in which users interact with the frontend interface through a web browser, while the backend server processes application requests, manages business logic, and communicates with the database.

The frontend of the application is developed using HTML, CSS, and JavaScript to provide an interactive and user-friendly interface. The backend is implemented using the Flask framework, which manages routing, authentication, session handling, product operations, messaging, and recommendation functionalities. SQLAlchemy ORM is used for secure and structured database interaction with SQLite database integration.

The system also incorporates advanced modules such as collaborative filtering for personalized product recommendations and location-based product discovery using distance calculation techniques. User interactions including product views, wishlist activity, and purchase history are analyzed to generate relevant recommendations. The application is deployed on a cloud-hosted environment using Gunicorn to ensure reliable accessibility and performance.

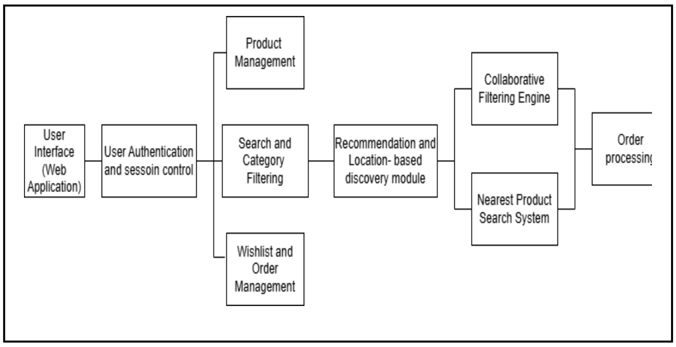

Fig. System Architecture

IV. METHODOLOGY

4.1 Content-Based Filtering

Content-based filtering is used to improve product recommendations by analyzing product attributes and user preferences. The system compares factors such as product category, description, price range, and keywords with user interaction history including wishlist activity, product views, and search behavior. Based on this comparison, the system identifies items that closely match user interests and generates personalized recommendations. This approach improves product relevance, reduces search effort, and enhances overall user engagement by continuously adapting to individual preferences within the platform.

4.2 Location-Based Tracking Algorithm

A location-based tracking algorithm is implemented to help users discover products available nearby within the campus environment. The system utilizes stored location data of users and sellers and applies distance calculation techniques to estimate proximity between them. Products are then ranked based on distance, giving priority to nearby listings for faster access and reduced travel time. This mechanism improves transaction efficiency, supports quick product exchange, and enhances user convenience by enabling location-aware product discovery within the campus marketplace.

V. RESULT AND OUTPUT

The proposed Campus Mart system successfully provides a secure and centralized platform for students to buy, sell, and exchange products within the campus community through features such as product listing, search, messaging, and order management.

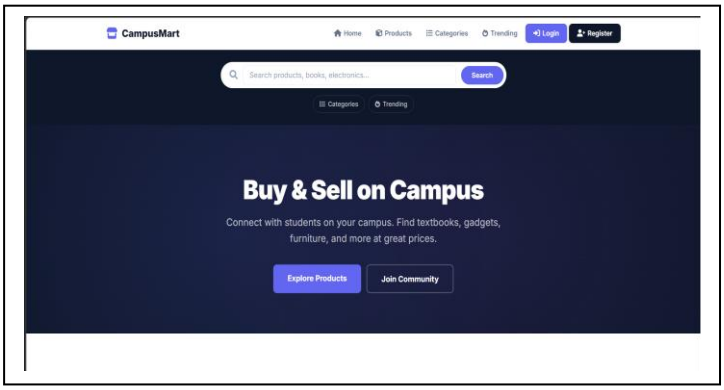

Fig. Homepage

The integration of collaborative filtering and location-based product discovery improves personalized recommendations, enhances product accessibility, and supports faster and more convenient campus transactions.

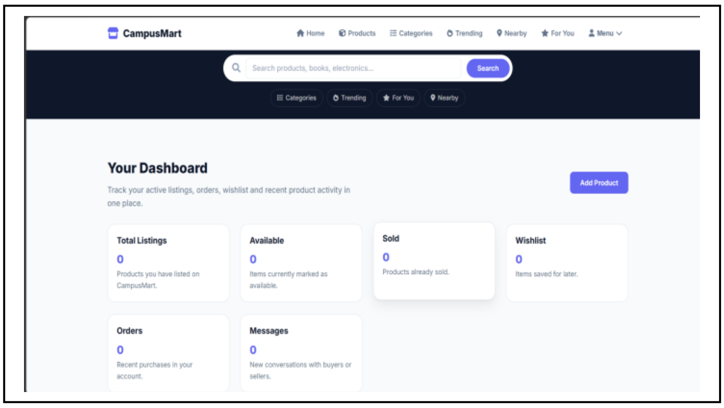

fig. Dashboard

VI. FUTURE SCOPE

- Real-time chat and notification systems can be added to improve communication and user engagement between buyers and sellers.
- Security and scalability can be enhanced through college email verification, AI-based fraud detection, mobile application support, and migration to advanced database systems like PostgreSQL.
- Advanced features such as hybrid recommendation systems, payment gateway integration, and image-based product search can be implemented to improve personalization, usability, and overall marketplace efficiency.

VII. CONCLUSION

The Campus Mart system provides a secure and efficient platform for campus-based buying, selling, and exchange of products within student communities. It addresses the limitations of informal trading by offering organized product management, secure authentication, direct buyer–seller communication, and improved product discovery. The use of collaborative filtering and location-based search enhances user experience through personalized recommendations and nearby product access. Developed using Flask, SQLAlchemy ORM, and cloud deployment, the system ensures reliable performance, scalability, and efficient data handling. Overall, it promotes affordable student commerce, encourages product reuse, and strengthens interaction within the campus ecosystem.

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